higher education & training

Department:
Higher Education and Training
REPUBLIC OF SOUTH AFRICA

NATIONAL CERTIFICATE (VOCATIONAL)

CIVIL AND STRUCTURAL STEEL WORK DETAILING

(First Paper)

NQF LEVEL 3

(12050663)

13 November 2019 (X-Paper)
09:00–11:00

REQUIREMENTS: **Answer book**
A3 Drawing paper and drawing instruments

Calculators may be used.

This question paper consists of 5 pages, 1 diagram sheet, 1 addendum and 1 answer sheet.

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TIME: 2 HOURS
MARKS: 75

INSTRUCTIONS AND INFORMATION

1. Answer ALL the questions.
2. Read ALL the questions carefully.
3. Number the answers according to the numbering system used in this question paper.
4. Start each question on a NEW page.
5. Rule off across the page on completion of each question.
6. Answer QUESTION 1.5 on the ANSWER SHEET (attached).
7. Write neatly and legibly.

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QUESTION 1: REINFORCED CONCRETE DETAILING

1.1 A bending schedule is also known as a bar schedule.

Name THREE advantages of a bar bending schedule. (3 × 2) (6)

1.2 Explain the following description for dimensioning reinforcing bars by copying and completing the table below in the ANSWER BOOK:

16 R12 – 05 –150T

16	
R	
12	
05	
150	
T	

(6 × ½) (3)

1.3 Draw the conventional symbols of the following abbreviations:

1.3.1 EW

1.3.2 ABR

1.3.3 STGD

(3 × 2) (6)

1.4 Name FOUR materials commonly used for formwork. (4)

1.5 Use the following information to complete the bending schedule for the steel reinforcement in the beams in a construction project on the ANSWER SHEET (attached).

Write your EXAMINATION NUMBER in the space provided, detach it from the question paper and hand it in with the ANSWER BOOK.

- The bars must be cut according to the shape code 32.
- Bars for six beams are needed. Each beam must have 4 high-yield steel bars of 16 mm and 6 plain 8 mm mild steel bars.
- Each of the 16 mm bars must have a straight length of 600 mm. The unbent length of these bars must be 900 mm.
- Each of the mild steel bars must have a straight length of 800 mm. The unbent lengths of these bars must be 1 200 mm.
- The high-yield bars have a bar mark of 25. The bar mark for the mild steel bars is 26.

Calculate the total bars for each type. (16)

[35]

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QUESTION 2: STRUCTURAL STEEL WORK DETAILING

2.1 Describe a lattice girder. (3)

2.2 Make a freehand drawing of a Warren-type lattice girder. (5)

2.3 Name TWO types of roof sheeting that are used. (2)

2.4 Name THREE types of bolts that are used for connecting structural work. (3)

2.5 Choose a description from COLUMN B to matches an item in COLUMN A. Write only the letter (A–J) next to the question number (2.5.1–2.5.8) in the ANSWER BOOK.

COLUMN A		COLUMN B	
2.5.1	Retaining wall	A	beam manufactured from plate material and used for large pans
2.5.2	Formwork	B	the main beam supporting the roof
2.5.3	Stirrup	C	a type of reinforcing in the beams that holds rebars in place and that resists shear
2.5.4	Rafter	D	temporary structure or mould into which the newly mixed concrete is placed and compacted to harden
2.5.5	Cleat	E	pieces of plate found at the intersection to facilitate a connection between
2.5.6	Girder	F	individual piece of timber, plywood or steel that is used to make formwork
2.5.7	Gusset plate	G	a short length of angle or channel section welded or bolted to a steel member to enable it to be fastened to another member
2.5.8	Purlin	H	freestanding wall that is used to hold back soil
		I	fixed or set securely or deeply into the cement or concrete
		J	the section used to support the roof sheeting

(8 × 1) (8)

2.6 On the DIAGRAM SHEET (attached) structural steel work is shown.

2.6.1 Which bolt must be fitted into hole C in the structural steel work shown? (1)

2.6.2 Name the parts indicated as A, B and D. Write only the answer next to the letter in the ANSWER BOOK. (3)
[25]

QUESTION 3: WELDING

3.1 Full penetration butt welds are formed when the parts are joined together within the thickness of the parent metal. For thin parts, it is possible to achieve full penetration of the weld. For thicker parts, edge preparation may have to be done to achieve the welding.

Name the various types of distortion and dimensional changes that may occur when welding. (5)

3.2 Refer again to the structural steel work shown on the DIAGRAM SHEET (attached).

3.2.1 Name the welding symbol shown at E.

3.2.2 Name the welding symbol shown at F.

(2 × 2) (4)

3.3 What metals can be weld with SMAW? (1)

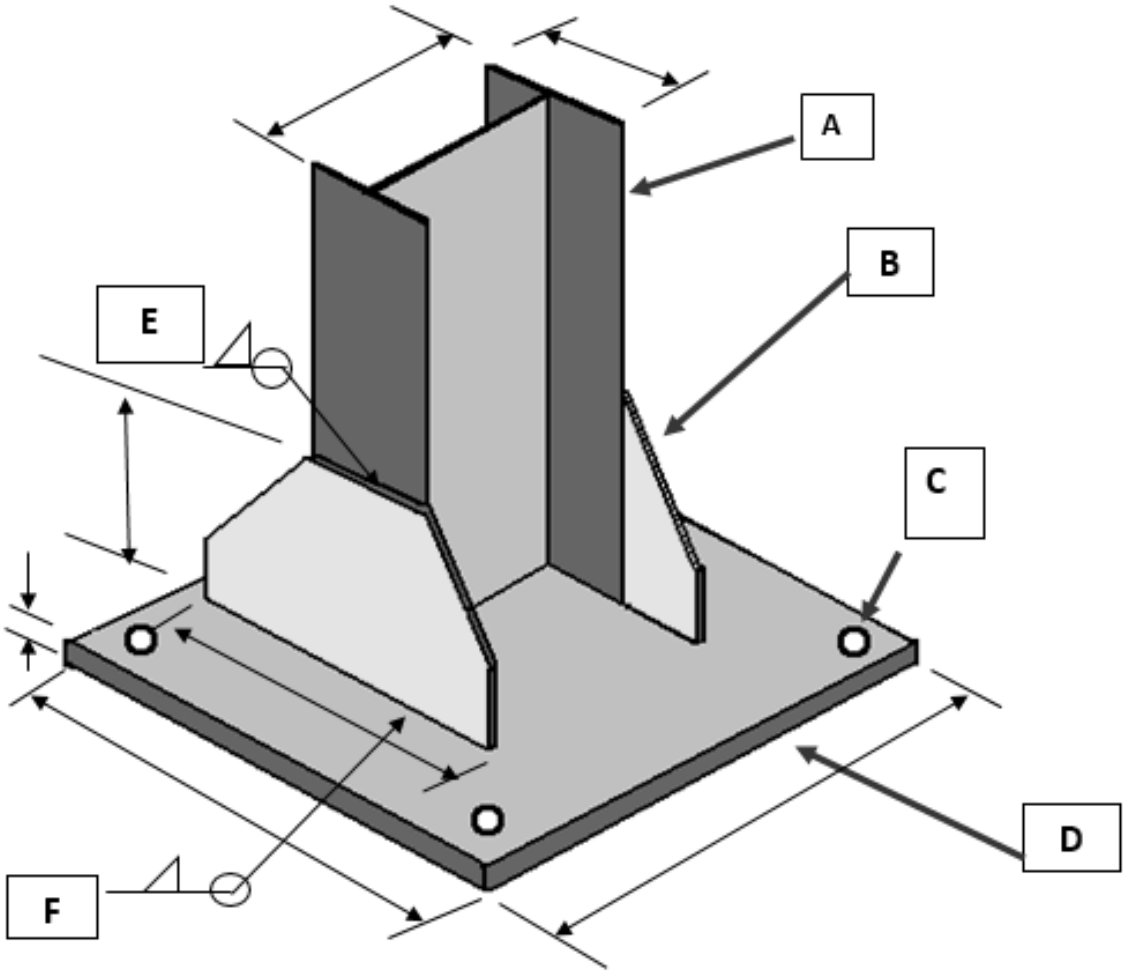
3.4 Name FIVE basic welding joints. (5)
[15]

TOTAL: 75

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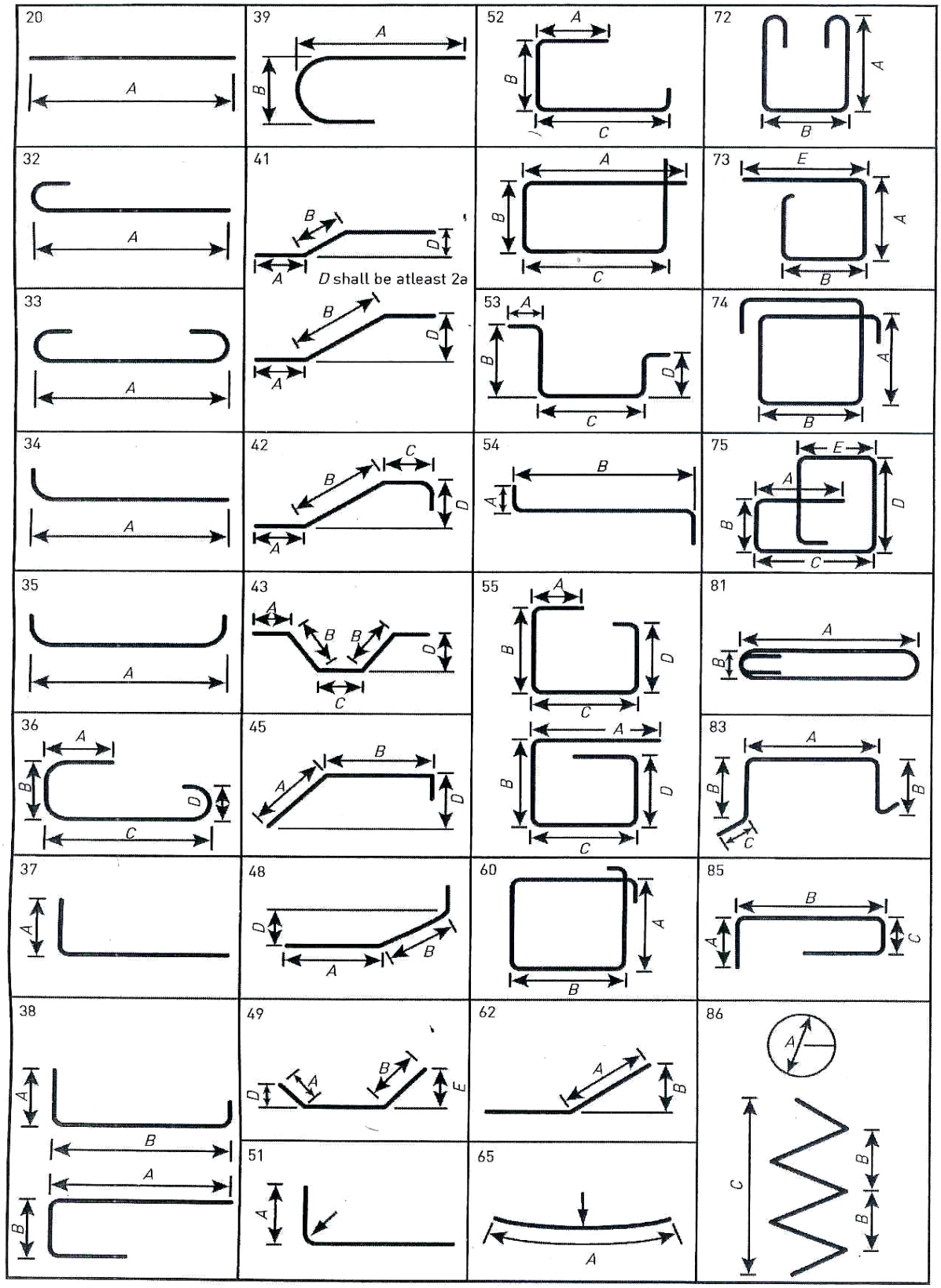
DIAGRAM SHEET



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ADDENDUM



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EXAMINATION NUMBER: | | | | | | | | | | | | | |

QUESTION 1.5

A	B	C	D	E	F	G	H				
Member								**Bending dimensions**			
	Bars per member	**No. of members**	**Total bars**	**Type/ diameter**	**Bar mark**	**Shape code**	**Total length**	**A**	**B**	**C**	**D**
Beam											
Beam											

(16)

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